ONE DISTRICT ONE PRODUCT
UTTAR PRADESH

A GUIDE TO PRODUCT QUALITY STANDARDS

SPORTS GOODS

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INDEX

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ABBREVATIONS

ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Co-operation
AS	Australian Standards
ASTM	American Society for Testing and Materials
BICON	Bio Security Import Condition System
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPSC	Consumer Products Safety Commission
CPSIA	Consumer Product Safety Improvement Act
CPSA	Consumer Product Safety Act
DOT	Department of Transportation
EC	Council Regulation (of the European Parliament and Council)
ECHA	European Chemicals Agency
EN	European Standards
EU	European Union
FDA	Food and Drug Administration
FIBA	Fédération Internationale de Basketball
FIFA	Fédération Internationale de Football Association
FHSA	Federal Hazardous Substances Act
GCC	Gulf Co-operation Council
GSO	GCC Standardization Organization
GPSD	General Product Safety Directive
ICF	International Carrom Federation
IITF	International Table Tennis Federation
IS	Indian Standard
ISI	Indian Standards Institution
ILAC	International Laboratory Accreditation Co-operation
ISO	International Organization for Standardization
MCC	Marylbone Cricket Club
MDF	Medium Density Fibre
MLA	Multilateral Recognition Arrangement

MRA	Mutual Recognition Arrangement
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RoHS	Restriction of Hazardous Substances Directive
SASO	Saudi Standards, Metrology and Quality Organization
SRL	Specific Release Limits
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This part of the compendium brings relevant regulations and standards pertaining to the Sports goods industry and goods produced in the state. Uttar Pradesh is the second largest producer of sports goods in India after Jalandhar. The sports goods industry is concentrated in and around Meerut which is a major source of employment not only in the segment of sporting goods but also in the various ancillary industries supporting the industry.

SPORTS GOODS, MEERUT

1. INTRODUCTION

Sports industry in India is nearly a century old and has flourished, driven by a skilled workforce. Being labour–intensive in nature, the industry provides employment to more than 500,000 people. India's sporting goods are popular around the world and have made a mark in the global sports goods market as products of high quality. The domestic industry exports nearly 60% of its total output.

Sporting equipment, also called sporting goods, are the tools, materials, apparel, and gear used to compete in a sport and varies as per the requirements of the sport. The equipment ranges widely from inflatable balls, bats, nets, to protective gear like helmets and guards. Over time, sporting equipment has evolved because sports have started to require more protective gear to prevent injuries to the competing sportsmen. Sporting equipment, now a day, may be found in any departmental store.

Historically, many sports players have developed their own sporting equipment over time. For instance, the use of a football dates back to ancient China, between 225 BC and 220 AD. As football is still one of the most popular sport in the 21st century, the material of the ball has completely changed over the centuries; from being made out of animal skin, to being lined with multiple layers of polyester or cotton.

As the sporting equipment industry improves, so does the performance of sportsmen. This is due to the fact that the equipment has become more efficient, lighter and stronger, forming a bio-mechanical system, which interacts with the athlete in real time and responds to the challenges on the playing field such as high performance running shoes or advanced monitors to track the physical health of the athlete.

Several enterprises including small and medium scale units in Meerut are engaged in producing world class sports goods. Meerut is known for manufacturing cricket and boxing equipment, badminton racquets, athletics and gymnastic equipment, swimwear, inflatable balls, hockey, table tennis equipment and skating equipment.

2. CRICKET BALL

It is generally believed that cricket originated as a children's game in the south-eastern counties of England, sometime during the medieval period. Although there are claims for prior dates, the earliest definite reference to cricket being played comes from evidence given at a court case in Guildford in January 1597 (Old Style), equating to January 1598 in the modern calendar.

Cricket is a bat-and-ball game played on a cricket field between two teams of eleven players each. The field is usually circular or oval in shape and the edge of the playing area is marked by a boundary, which may be a fence, part of the stands, a rope, a painted line or a combination of these; the boundary must if possible be marked along its entire length.

A cricket ball is a hard, solid ball used to play cricket and its manufacture is regulated by cricket law at first-class level. The trajectory of a cricket ball when bowled, through movement in the air, and off the ground, is influenced by the action of the bowler and the condition of the ball and the pitch, while working on the cricket ball to obtain optimal condition is a key role of the fielding side. The principal method through which the batsman scores run, is by hitting the ball, with the bat, into a position where it would be safe to take a run, or by directing the ball through or over the boundary. Cricket balls are harder and heavier than baseballs.

2.1. Raw Materials

Leather, Portuguese Bark and String

2.2. Manufacturing Process

A cricket ball basically is made with a core of cork, which is layered with tightly wound string, and covered by a leather case with a slightly raised sewn seam.

a. **Outer Layer:** The outer layer of a cricket ball is a strong casing constructed of leather. Both red and white balls are covered in the same type of leather casing, however, the colour of the dye used to finish the leather is different. The quality of leather impacts the overall quality of the ball. The highest quality balls feature highest-grade leather which is hand-cut, tanned and stitched.

b. **Inner Layer:** Underneath the leather shell, multiple materials are combined to create a solid inner core. The core is a golf ball sized combination of cork and rubber. An additional layer of cork is then applied to the core, which is then wrapped in yarn. This is repeated up to five times (depending on the size of the ball). Finally, a plastic cap covers the layers to protect the inner part from any moisture.

In general, the cricket balls are typically made from the core up. The process starts with treated Portuguese bark, which is compressed by a machine into a golf ball sized cork core. The core is then coated in two precisely measured additional layers of cork. The enhanced core is spun in five layers of high-quality yarn. This forms the quilt and gives the ball an even bounce.

The tough outer casing starts with two high-quality leather sheets which are cut into ovals – each representing a quarter of a ball. Two sections are then stitched together, forming a perfect hemisphere. These half balls are not yet joined together, but a double seam is sewn around each hemisphere to add strength.

Figure 1: Installing the core and weight check

Two hemispheres are hand-stitched together around the core, using high-quality linen thread. In a top-quality ball suitable for highest levels of competition, the covering is constructed of four pieces of leather shaped similar to the peel of a quartered orange, but one hemisphere is rotated by 90 degrees with respect to the other. The "equator" of the ball is stitched with string to form the ball's prominent seam, with six rows of stitches. The remaining two joints between the leather pieces are stitched internally forming the quarter seam. When formed, the balls are finished with a spray of nitrocellulose lacquer. This adds strength and shine to the now finished cricket ball.

Figure 2: Hand stitching

2.3. General Quality Checks

Following are a few visual quality parameters/checks:

a. The weight of the ball should be within the specified ranges
   - Men's Cricket: 156 gm. – 163 gm.
   - Women's Cricket: 140gm – 151 gm.

b. Stitching should be uniform and not damaged

c. The polish should be uniform

3. CRICKET BAT

A cricket bat is a specialised piece of equipment used by the batsmen in the sport of cricket to hit the ball, typically consisting of a cane handle attached to a flat-fronted willow-wood blade. It may also be used by a batter who is making their ground to avoid a run out, if they hold the bat and touch the ground with it. The length of the bat may be no more than 38 inches (96.5 cm) and the width no more than

4.25 inches (10.8 cm). Its use is first mentioned in 1624. Since 1979, a law change stipulated that bat can only be made from wood and therefore, the bat is now made up of wood only. The wood used is from the Kashmir or English willow tree. Bats of any other materials or a mixture of metals is not permitted.

3.1. Raw Materials

Willow Wood, Linseed Oil, Cane and Rubber grip for the handle

3.2. Manufacturing Process of making a cricket bat

The bat has a long handle and one side has a smooth face. The blade of a cricket bat is a wooden block that is generally flat on the striking face and with a ridge on the reverse (back) which concentrates wood in the middle where the ball is generally hit.

a. **Selection of Wood:** The bat is traditionally made from willow wood, specifically from a variety of white willow called cricket bat willow (*Salix alba var. caerulea*), treated with raw (unboiled) linseed oil, which has a protective function. This variety of willow is used as it is very tough and shock-resistant and does not dent nor splinter on the impact of a cricket ball at a high speed, while also being light in weight. The wooden *clefts* (wood cut in a wedge shape for making bats) are cured for six months.

Figure 3: Seasoning of wood

b. **Fabricating the blade:** The striking face of the bat is called the blade. Using a plane, the bat is given its shape as per the specifications laid down for cricket bats.

Figure 4: Fabricating the blade

c. **Pressing:** To make the bat harder and compress the wood. The mechanical process of pressing bat is very popular and is widely used in the making of bats. Different kinds of roller are used to apply pressure by hydraulic, pneumatic and weighted cantilever systems. This pressure is increased incrementally up to 2.5 tons per square inch.

Figure 5: Pressing the blade

d. Fixing the handle: The blade is connected to a long cylindrical cane handle, similar to that of a mid-20th-century tennis racquet, by means of a splice. The handle is usually covered with a rubber grip. Bats incorporate a wooden spring design where the handle meets the blade. The current design of a cane handle spliced into a willow blade through a tapered splice was the invention in the 1880s of Charles Richardson, a pupil of Brunel and the first Chief Engineer of the Severn Railway Tunnel. Spliced handles had been used before this but tended to break at the corner of the joint. The taper provides a more gradual transfer of load from the bat's blade to the handle and avoids this problem. The edges of the blade closest to the handle are known as the shoulders of the bat, and the bottom of the blade is known as the toe of the bat.

Figure 6: Fixing the handle

e. Knocking: The bat is prepared for use by knocking it in. This is done using special machines which repeatedly strike the front part of the blade repeatedly to match the impact of a cricket ball. This force is gradually increased to emulate the bowling speeds.

Figure 7: Knocking the bat

f. Installing the grip: Lastly the grip is fixed and the stickers and decals (a picture,name or logo made to be transferred to a surface)applied.

3.3. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for damages to the cleft such as knots.
- Check for warpage and minor cracks in the wood.
- The sizes should conform to standard measurement laid down by the ICC (maximum dimensions of a cricket bat will be 108 mm in width, 67mm in depth with 40mm edges)

4. CARROM BOARD

Carrom is a table top game of Indian origin. The game is very popular in South Asia, and is known by various names in different languages. In South Asia, many clubs and cafés hold regular tournaments. Carrom is very commonly played by families, including children, and at social functions. Carrom is not a patented game. Different standards and rules exist in different areas. It became very popular in the United Kingdom and the Commonwealth during the early 20th century.

The International Carrom Federation (ICF) was formed in the year 1988 in Chennai, India. The formal rules for the Indian version of the game were published in 1988. In the same year the ICF officially codified the rules.

4.1. Raw Material

Hardwood, plywood, Wood Glue, Stain & Polyurethane clear coat

4.2. Manufacturing Process

A Carrom board is a square smooth flat wooden board that can be 72cm or 74cm square. In each corner is a circular hole that can be 51mm in diameter and underneath each hole is a net to catch the pieces in a similar way to a snooker table. The dimensions of the standardized game are a 29 square inches (74 cm) playing surface on a board of lacquered plywood. The playing surface of a carrom board is normally made of birch plywood and it is at least 8 mm thick with a completely smooth surface. The basic process of making a carom board is:

- The frame is made by cutting the hardwood edge pieces and using a lap joint and gluing them together. To allow the glue to set firmly, clamps are used to keep the frame immobile
- Plywood is cut to the size desired and the pockets are cut out along the four edges. The plywood is sanded and stained.
- The designs and artwork are made on the surface and after final markings are completed, poly urethane polish is used to give a smooth surface.
- Adhesive is applied along the edges and the frame is attached to it.
- The Carrom men, Queen and the Striker are all cut in circular shape from wood. The average thickness varies from 7 mm to 9 mm and should have a mass of

5 to 5.5 gms. The carrom men are designed to slide when struck by the striker. The striker is larger than the carrom men having a diameter of around 40 mm to 50 mm.

Figure 8: Marking done on Plywood Surface

Figure 9: Final Finishing

4.3. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for warpage of plywood
- The surface should be smooth to enable the carrom men to slide on the surface
- The sizes should conform to standard measurement laid down by the International Carrom Federation (ICF)

5. TABLE TENNIS

Table Tennis, also known as Ping-Pong and whiff-whaff, is a sport in which two or four players hit a lightweight ball, also known as the Ping-Pong ball, back and forth across a table using small rackets. All equipment used in this sport are regulated by the International Table Tennis Federation (IITF). The main components used in this game are as follows:

- Table
- Ball
- Racket
- Nets

5.1. Table Tennis Table

The official sized table is 2.74 m (9.0 ft.) long, 1.525 m (5.0 ft.) wide, and 76 cm (2.5 ft.) high that can yield a uniform bounce of about 23 cm (9.1 in) when a standard ball is dropped onto it from a height of 30 cm (11.8 in), or about 77%. The table or playing surface is uniformly dark coloured and matte, divided into two halves by a net at 15.25 cm (6.0 in) in height. The ITTF approves only wooden tables or their derivatives. Concrete tables with a steel net or a solid concrete partition are sometimes available in outside public spaces, such as parks but are not approved and rather are meant for outdoor uses.

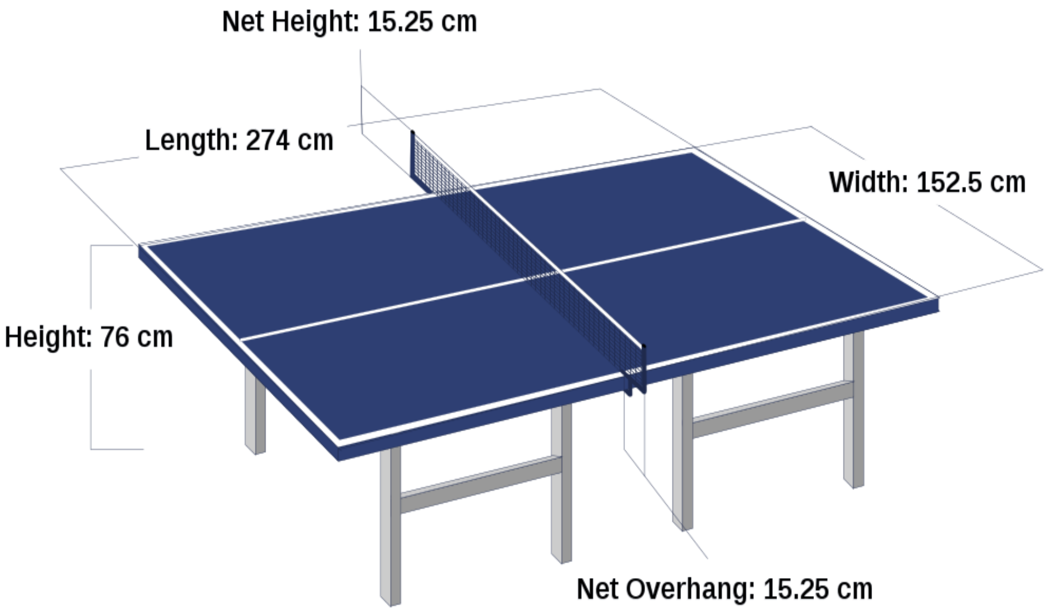

Figure 10: Table Tennis Table dimensions

a. **Raw materials:** The raw materials used in the construction of the table are as:
   - Powder coated heavy gauge hollow steel sections (Rectangular or tubular)
   - Medium Density Fibre (MDF) board/wood
   - Paint
   - Net

b. **Table Tennis Table manufacturing process**
   - **Fabricating the frame:** The steel sections are cut as per the dimensions defined by the governing body (ITTF). The sections are joined together using either screws or welded together. The table frame should be sturdy enough to prevent any movement of the table during use. If casters are being added to the legs to facilitate easier movement they should have a locking mechanism.
   - **Table Top:** Medium Density Fibre (MDF) is usually preferred for making of the table tops as it provides the required bounce to the ball. The table top is usually between 25 mm to 30 mm thick.
   - **Painting:** The table top has to be painted as directed by the International Table Tennis Federation (IITF). For this purpose, Alkyd paint or oil-based paints having a 60-degree specular gloss are used as they provide a smooth and uniform surface and are extremely durable.

The colours used are green or blue.

- **Net:** It is usually made up of synthetic plastic, nylon mesh, or cotton cloth. It can be affixed to a table via the post clamps quickly.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- Dimensions should be as prescribed by the governing bodies
- The surface should be free from damages and should be smooth
- The table should not wobble and should be sturdy
- Colour should be uniform with no patches and in an even layer

5.2. Ball

The international rules specify that the game is played with a sphere having a mass of 2.7 grams (0.095 oz.) and a diameter of 40 millimetres (1.57 in). The rules say that the ball shall bounce up 24–26 cm (9.4–10.2 in) when dropped from a height of 30.5 cm (12.0 in) onto a standard steel block thereby having a coefficient of restitution of 0.89 to 0.92. Balls are now made of a polymer instead of celluloid as of 2015, coloured white or orange, with a matte finish.

Manufacturers often indicate the quality of the ball with a star rating system, usually from one to three, three being the highest grade. As this system is not standard across manufacturers, the only way a ball may be used in official competition is upon ITTF approval (the ITTF approval can be seen printed on the ball).

a. **Raw Materials:** Cellulose Nitrate, Cellulose Acetate, Table Tennis Rubber, Wood, Adhesives, Medium density fibre board (MDF)

b. **Manufacturing:** Traditionally, the ball was made of celluloid. In 2014, their use was prohibited by IITF. They are now made of cellulose acetate.

- The process starts with circular plastic sheets, where individual plastic sheets are checked for quality and weighed.
- The balls are made from two separate ball halves. During the first stage, round copper heads press the sheets into shape. Warm water is also dripped continuously on the plastic sheets to prevent cracks.
- The edges are trimmed using machines and the ball halves are glued to each other.

- The assembled balls are given the final finish and stored for up to 15 days in temperature and humidity-controlled environment.
- Automated machines check for the final quality. Any ball which is not found up to standards is discarded. For competition purposes, a second visual check by hand is carried out.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- The ball should not have defects like uneven seams
- The balls should weigh 2.68 gm. – 2.76 gm.
- The balls are checked for size and should be in the range of 40.00 mm – 40.40 mm

5.3. Racket

A table tennis racket (also known as a "paddle" or "bat") is used by table tennis players. It is usually made from laminated wood covered with rubber on one or two sides depending on the player's grip. Unlike a conventional "racket", it does not include strings strung across an open frame. In USA, the generally used term for racket is "paddle" while Europeans and Asians use the term "bat" and the official ITTF term is "racket".

- IITF regulations allows different sizes but the blade must be flat and rigid. Also, it should comprise of 85% natural wood. However, most rackets are usually 6 inches wide and 10 inches long. (Including the handle.)
- Rubber sheets are attached to each side of the blade. Modern table tennis rubber is usually composed of two layers: a layer of foam underneath and a layer of actual rubber on the surface. There are four common types of table tennis rubbers: short pips, long pips, antispin, and inverted. The thickness and density of the sponge layer underneath also affects how the rubber will handle the ball. Because of the large variety of table tennis rubbers, there are a larger variety of playing styles in table tennis than in other racket sports.

Figure 11: Table tennis rubber

- The Individual players' style determines the type of rubber which gets applied to the racket.

Usually both sides may carry different rubbers.

- **Inverted:** Inverted rubber is an offensive rubber and the most popular type of rubber. They are smooth on the outside, leaving the pips on the inside touching the foam, which is why it is called inverted. Inverted rubber's wide variety of speeds and spins make it a favourite among players. Manufacturers will provide speed and spin ratings to differentiate their type of inverted rubber from others. The faster the speed rating, the harder it is to control.
- **Short pips:** Short pips rubber, also called pips out rubber, is a more controlled attack rubber that provides more speed but less spin to an attack than inverted rubber. Due to the small contact surface with the ball, short pips rubber is not easily affected by the opponent's spin; instead, it will knock a ball away. Therefore, the rubber is usually used for blocking, hitting, and counterattack strokes.
- **Long pips:** Long pips rubber is a defensive rubber. As its name suggests, the rubber has longer pips than short pips, and usually it also has thinner pips. The flexibility of the pips creates a low friction effect. As a result, it provides little spin because the spinning ball slips on the rubber.
- **Anti-spin:** Anti- spin rubber is a defensive rubber with a smooth surface and very soft sponge. Due to the soft sponge, this rubber dampens speed of the ball and returns the ball at a low speed. It is usually used by defensive players on one side of their racket only. Unlike inverted rubber, it does not grip the ball, so opponents' balls are returned with little spin. However, against heavy spin, anti- spin rubber will return the ball with an opposite spin, similar to long pips.

The two main grip styles are shake hand and pen hold.

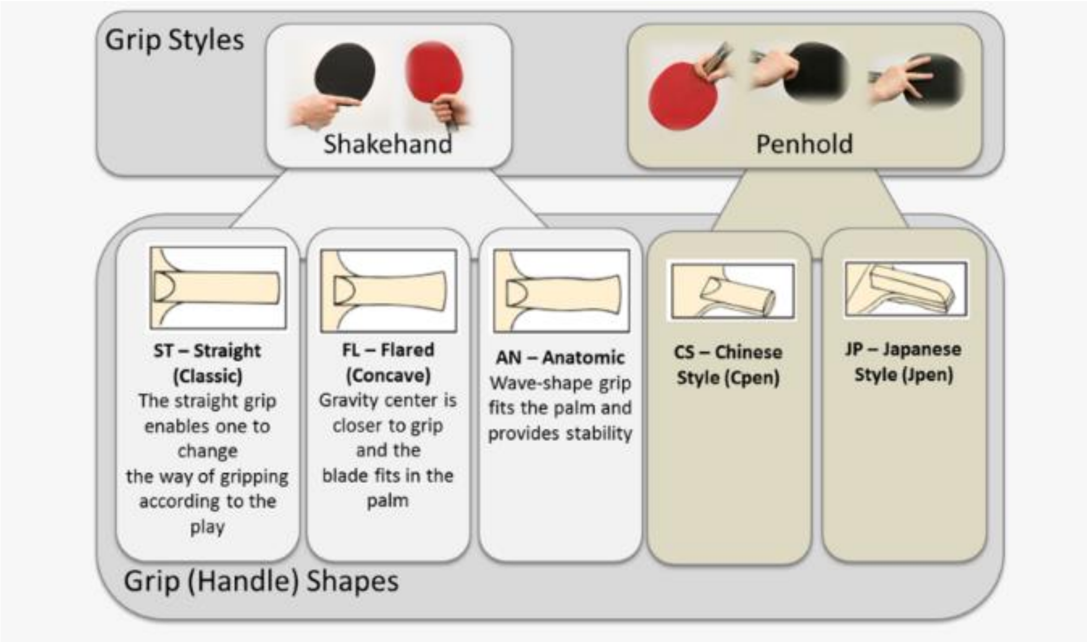

Figure 12: Table tennis racquet grip

a. **Raw Materials:** Cellulose Nitrate, Cellulose Acetate, Table Tennis Rubber, Wood, Adhesives, Medium density fibre board (MDF)

b. **Manufacturing of Table tennis rackets**

- **Blade making:** The making of the racket starts with the blade. The blade is made up of up to as many as seven layers of wood, cork, glass fibre, carbon fibre. These are glued together and pressed using a mechanical press.
- **Cutting of the shape:** The shape of the racket is drawn out on the compressed sheet and cut out. The edges and surfaces are carefully sanded to ensure a smooth finish. A surface lipping usually of veneer is applied to the edges of the racket to prevent damage to the exposed edges of the laminated layers
- **Making the handle:** The handle of the racket is fashioned out of two identical pieces of hardwood. They are shaped into a bevelled shape using woodworking tools. These are then attached to the racket.

Figure 13: Fixing the handle

- **Attaching the rubber:** Depending on the choice of the rubbers by the player in case of a customized racket, the rubbers are selected. Modern table tennis rubber is usually composed of two layers: a layer of foam underneath and a layer of actual rubber on the surface. They are glued onto the blade faces. As per current rules one side of the paddle has to be black and the other side can be blue, red, pink, green or purple.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- Check for proper adhesion of rubber to the blade
- Check for any cuts, wear and tear of the rubber
- Check for any damages and warping in the racket

6. INFLATABLE BALLS

A ball is a round object (usually spherical, but can sometimes be ovoid) with various uses. It is used in ball games, where the play of the game follows the state of the ball as it is hit, kicked or thrown by players. Balls can also be used for simpler activities; such as catch or juggling.

One of several advantages of an inflatable ball is that it can be stored in a small space when not inflated, since inflatables depend on the presence of a gas to maintain their size and shape. Function fulfilment per mass used compared with non-inflatable strategies is a key advantage.

6.1. Football/Soccer Ball

Kicking ball games arose independently across multiple cultures. The Chinese competitive game cuju (literally "kick ball") resembles modern association football. Cuju players could use any part of the body apart from hands and the intent was kicking a ball through an opening into a net.

The football/soccer ball comes in different sizes as given below:

Chart showing sizes and types of balls (For ages 3 and up)

S. No.	Age	Ball Type	Size (In Inches)	Weight (In Grams)
1	3 Years and Under	Size 1	18-20	200
2	3 to 5 Years	Size 2	20-22	250
3	5 to 8 Years	Size 3	23-24	300-320
4	8 to 11 Years	Size 4	25-26	350-390
5	12 Years and Above	Size 5	27-28	410-450

** Size 5 is the size used in international competitions*

a. Raw Materials: The raw materials used for manufacturing a football are mainly:

- Polyurethane
- Butyl rubber
- Polyester threads
- Laminated cotton sheets or polyester

b. Manufacturing

- Within a soccer ball is a bladder that holds air. The bladders are either made from latex or butyl, depending on the soccer ball quality. Butyl bladders are the most expensive, compared to balls using latex but can hold air longer. In order for the bladder to be made, latex or butyl rubber is heated carefully, and then poured into mould. When cooled, the round, flat bladder will be taken out and set aside.
- The soccer ball's bladder is bordered by lining layers. Such layers are comprised of polyester or laminated cotton sheets. There could be up to four layers in a soccer ball. If there are more layers, the ball will bounce higher. The most expensive balls however have four layers at least. The lining layers are affixed to the outer ball cover.
- The outside of the soccer balls or the outer cover is made from synthetic leather or rubber. Professional balls are usually made with synthetic leather while the practice balls are typically made of rubber.
- The material of the cover is stitched into the layers and the entire thing is sent into a machine that cuts out each panel along with the convenient holes from the stitch. If the ball has graphics, the machine will add those into the sheets of fabric before cutting it. Then the panels are finally glued or sewn together. Balls that are of the highest grade are hand-stitched with special cord that consists of five polyester lines that are all twisted together. The medium grade ones are stitched together with machines while those that are most inexpensive are glued together.
- The panels are then sewn together turning the ball inside out, the round bladder taken out of the mould and cooled off is sewn together

as well into sphere and inflated partially. When the panels are almost complete, the ball is be turned right side out and the bladder will be stuffed inside the ball. The rest of the panels are then sewn closed around its inflating spot and then the bladder is be inflated. The final steps are done all by hand.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- For official matches check the size and weight of the football as prescribed
- Check for proper stitches and punctures if any.
- When dropped from a height of 2.0m under test conditions, it should bounce back to a height of at least 1.35m
- The paint should be uniform and not flaking off

6.2. Basket Ball

Basketball is a team sport in which two teams, most commonly of five players each, opposing one another on a rectangular court, compete with the primary objective of shooting a basketball (approximately 9.4 inches (24 cm) in diameter) through the defender's hoop (a basket 18 inches (46 cm) in diameter mounted 10 feet (3.048 m) high to a backboard at each end of the court) while preventing the opposing team from shooting through their own hoop.

The Basketballs come in various sizes as below:

Chart showing sizes and types of balls

S. No.	Age	Ball Type	Size (In Inches)	Weight (In Ounces)
1	0-4 Years and Under	Nerf Toy	9-20	1-5
2	2 to 4 Years (Boys & Girls)	Size 1	16	8
3	4 to 8 Years (Boys & Girls)	Size 3	22	10
4	5 to 8 Years (Boys & Girls)	Size 4	25.5	14
5	9 to 11 Years (Boys & Girls)	Size 5	27.5	17
6	12-14 Years (Boys) and ages 12 and up (Girls)	Size 6	28.5	20
7	15 Years and above (Men & Boys)	Size 7	29.5	22

** Size 7 is the size used in international competitions*

a. **Raw Materials:** The raw materials used for manufacturing a basketball are:
   - Leather
   - Composite/Synthetic Leather
   - Rubber
   - Laminated cotton sheets or polyester
   - Butyl rubber

b. **Manufacturing Process**
   - **Bladder:** Within a basketball is a bladder that holds air. The bladders are made from butyl rubber. The butyl rubber is melted into flat panels that are then attached together into the shape of a ball.
   - **Shaping the balls insides:** While the bladder is round, it's relatively misshaped. In order for basketballs to share a uniformed shape and size, they must be shaped with the help of a machine, which wraps polyester or nylon threads around the inner bladder. The threads are what create the spherical shape of the ball and prevent it from becoming deformed. Street balls typically feature polyester thread, while professional balls are shaped with nylon thread.
   - **Cover of the Ball:** A basketball's cover can be made from a variety of materials. The highest-quality balls are covered in quality leather. Other materials include synthetic rubber, natural rubber or composition. The material is unrolled and then cut into six separate panels that will eventually come together to wrap around the ball. Before the panels are placed on the ball, however, any embossed or stamped markings are placed onto the panels. If the panels are leather, they are then stitched together around the ball. Synthetic rubber, natural rubber and composition panels are held on with glue. Decals (a picture, name or logo made to be transferred to a surface), information and foil markings are applied by hand after the panels have been sealed onto the ball. The balls are then inflated and set aside for another 24 hours then inspected to see if they continue to hold air.

c. **General Quality Checks:** Following are a few visual quality parameters/checks:

- For official matches check the size and weight of the football as prescribed
- Check for proper stitches
- Check for wear & tear, if any
- Check for punctures, if any
- When dropped from 72 inches, the ball must rebound up to 52 to 56 inches.

7. REGULATORY BODIES AND CUSTODIANS OF THE LAW

Product/ Sport	International Body responsible for standards	Scope
Cricket bat	Marylebone Cricket Club (MCC)	Although the International Cricket Council is the global Governing Body for cricket, it still relies on MCC to write and interpret the Laws of Cricket. MCC's Laws sub-committee are responsible for the, decision making and drafting of the Laws and standards of cricket.
Cricket Ball	Marylebone Cricket Club (MCC)	Although the International Cricket Council is the global Governing Body for cricket, it still relies on MCC to write and interpret the Laws of Cricket. MCC's Laws sub-committee are responsible for the, decision making and drafting of the Laws and standards of cricket.
Table Tennis	International Table Tennis Federation	The rules & regulations of the ITTF can be found in the ITTF Handbook of 2021
Carrom Board	International Carrom Foundation	The ICF is tasked with compiling and enforcing rules and regulations governing Carrom and to ensure in all competitions, whether sanctioned by the ICF, or a Member, that such rules and regulations shall be applied in accordance with their terms.

8. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

8.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	NA	NA	NA

8.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	**a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations **b.** Regulation (EU) No. 995/2010 (EUTR)	**a.** European Commission **b.** European Chemical Agency (ECHA)	**a.** Hazardous chemicals **b.** Persistent organic pollutants **c.** General product safety directive
2	United States of America	**a.** Consumer Product Safety Improvement Act (CPSIA), Act of 2008 **b.** Consumer Product Safety Act (CPSA) Act of 1972 **c.** Federal Hazardous Substances Act (FHSA), Act of 1960.	**a.** Consumer Product Safety Commission (CPSC)	**a.** Regulations on chemicals **b.** Quantity of regulated material **c.** Product labelling
3	Australia	**a.** Biosafety Import Conditions (BICON) **b.** Illegal Logging Prohibition Act (2012) **c.** Illegal Logging Prohibition Regulation (2012)	**a.** Department of Agriculture, Water and Environment	**a.** Regulation of imports **b.** Treatment methods for wooden products **c.** Illegal Logging and prevention of materials obtained thus

9. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 707: 2011 (Reaffirmed Year: 2021)**	Timber technology and utilization of wood, bamboo and cane - Glossary of terms	USA	**ASTM D3853 – 12 (2017)**	Standard terminology relating to rubber and rubber lattices - Abbreviations for chemicals used in compounding
	IS 1141: 1993 (Reaffirmed Year: 2020)	Seasoning of timber – Code of practice	USA	**ASTM D1517 - 18**	Standard terminology relating to leather
	IS 1640: 2007 (Reaffirmed Year: 2017)	Glossary of terms relating to hides, skins and leather	Australia	**ISO 15115: 2019**	Leather - Vocabulary

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
Raw Materials (Refer Table 2.a & 2.b)	**IS 4422: 1985 (Reaffirmed 2020)**	Willow clefts for cricket bats	European Union	**BS EN ISO 4044: 2017**	Leather**:** Chemical tests. preparation of chemical test samples
	IS 1708: Part 1 to 18: 1986 (Reaffirmed Year: 2020)	Methods of testing of small clear specimens of timber	Middle East	**GSO ISO 10195: 2021**	Leather — Chemical determination of chromium(vi) content in leather — Thermal pre-ageing of leather and determination of hexavalent chromium
	IS 4144: 1967 (Reaffirmed Year: 2017)	Carrom strikers	Middle East	**GSO ISO 1656:2015**	Rubber, raw natural, and rubber latex, natural - Determination of nitrogen content
	IS 2179: 1979 (Reaffirmed Year: 2020)	Carrom board	Middle East	**GSO ISO 1598: 2015**	Plastics - Cellulose acetate - Determination of insoluble particles
	IS 3659: 1993 (Reaffirmed Year: 2017): 1993 (Reaffirmed Year: 2020)	Table tennis balls	-	**-**	-
	IS 4141: 1988 (Reaffirmed Year: 2017)	Table tennis rackets	-	**-**	-
	IS 4979: 1993 (Reaffirmed Year: 2017)	Table tennis table	-	**-**	-
	IS 4553: 1983 (Reaffirmed Year: 2017)	Leather for cricket ball	-	**-**	-
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 417: Part 1: 2003 (Reaffirmed Year: 2017)**	Footballs, volleyballs, basketballs, netballs, throw balls and water-polo Balls - Part 1 **:** Footballs	European Union	**BS EN 14468-1: 2015**	Table tennis Table tennis tables, functional and safety requirements, test methods
	IS 417: Part 3: 1986 (Reaffirmed Year: 2017)	Footballs, volleyballs, basketballs, netballs, throw balls and water-polo balls - Part 3 **:** Basketballs	European Union	**BS EN 14468-2: 2015**	Table tennis posts for net assemblies. Requirements and test methods
	IS 10800: 1983 (Reaffirmed Year: 2017)	Cricket balls	-	**-**	-
	IS 4143: 1967 (Reaffirmed Year: 2017)	Carrom-draughts	-	**-**	-
	IS 3659: 1993 (Reaffirmed Year: 2017)	Table tennis balls-specifications	-	**-**	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
Finishing (Refer Table 4.a & 4.b)	**IS 4905: 2015 (Reaffirmed Year: 2020)**	Methods for random sampling	European Union	**BS EN 1270**	Playing field equipment - Basketball equipment - Functional and safety requirements, test methods
	IS 4144: 1967 (Reaffirmed Year: 2017)	Carrom strikers	USA	**ASTM F2117 – 10 (2017)**	Standard test method for vertical rebound characteristics of sports surface/ball systems; acoustical measurement

9.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 707: 2011 (Reaffirmed Year: 2021)** - Timber technology and utilization of wood, bamboo and cane - Glossary of terms [Adjunct to IS 4422: 1985 (Reaffirmed 2020)]	▪ Glossary of terms
	IS 1141: 1993 (Reaffirmed Year: 2020) - Seasoning of timber – Code of practice [Adjunct to IS 4422: 1985 (Reaffirmed 2020)]	▪ Classification of timber for seasoning purposes ▪ Standards referenced ▪ Identification of defects
	IS 1640: 2007 (Reaffirmed Year: 2017) - Glossary of terms relating to hides, skins and leather	▪ Glossary of terms
	ASTM D1517 – 18 - Standard terminology relating to leather	▪ Glossary of terms
	ASTM D3853 - 12(2017) - Standard terminology relating to rubber and rubber lattices—Abbreviations for chemicals used in compounding	▪ Compilation of abbreviations for accelerators, vulcanizing agents, activators, antidegradants, plasticizers, softeners, processing aids, blowing agents, retarders, isocyanates, peroxides, and antireversion agents used in the compounding of rubber products
	ISO 15115-2019 - Leather - Vocabulary	▪ Glossary of terms

9.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 4422: 1985 (Reaffirmed 2020)** - Willow clefts for cricket bats	▪ Specification of species of timber ▪ Adjunct standards ▪ Seasoning requirements, moisture content ▪ Mechanical strength requirements
	IS 1708 (Part 1 to 18) - Methods of testing of small clear specimens of timber	▪ Methods of tests for determining moisture content of timber ▪ Determination of specific gravity, volumetric shrinkage, radial and tangential shrinkage and fibre saturation point, static bending strength, static bending strength under two-point loading, impact bending strength, compressive strength parallel to grain, compressive strength perpendicular to grain, hardness under static indentation, shear strength parallel to grain, tensile strength parallel to grain, tensile strength perpendicular to grain ▪ Test methods

Stage	Code	Scope
	IS 4553: 1983 (Reaffirmed Year: 2017) - Specification for leather for cricket balls (Adjunct to IS 10800: 1983 (Reaffirmed Year: 2017))	▪ Requirements of leather for cricket balls ▪ Methods and tests ▪ Adjunct standards ▪ Types of Leather basis methods of tanning
	IS 4144: 1967 (Reaffirmed Year: 2017) - Carrom-strikers	▪ Material requirements of carrom-strikers. ▪ Dimensional, and constructional requirements of carrom-Strikers. ▪ Constructional requirements of carrom-strikers.
	IS 707: 2011 (Reaffirmed Year: 2021) - Timber technology and utilization of wood, bamboo and cane - Glossary of Terms	▪ Definitions and glossary of terms
	IS 2179: 1979 (Reaffirmed Year: 2020) - Carrom board	▪ Grades of carrom boards ▪ Adjunct standards and referenced documents ▪ Dimensional details to be adhered to.
	IS 4143: 1967 (Reaffirmed Year: 2017) - Carrom draughts	▪ Dimensional and constructional requirements of carrom-draughts.(shape, dimensions and weight) ▪ Moisture content in the type of wood required for making.

9.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Materials	**BS EN ISO 4044-2017** – Leather - Chemical tests, preparation of chemical test samples	▪ Testing methodology ▪ Preparation of samples
	GSO ISO 10195-2021 - Leather — Chemical determination of chromium(vi) content in leather — Thermal pre-ageing of leather and determination of hexavalent chromium	▪ Test methodology to determine hexavalent chromium
	GSO ISO 1656-2015 - Rubber, raw natural, and rubber latex, natural - Determination of nitrogen content	▪ Macro-method and a semi-micro method for the determination of nitrogen in raw natural rubber and in natural rubber latex using variants of the Kjeldahl process.
	GSO ISO 1656-2015 - Rubber, raw natural, and rubber latex, natural - Determination of nitrogen content	▪ Method for the determination of the number of visible particles (including all kinds of contamination and black dirt) in cellulose acetate

9.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 417: Part 1: 2003 (Reaffirmed Year: 2017)** - Footballs, volleyballs, basketballs, netballs, throw balls and water-polo Balls - Part 1 : Footballs	▪ Methods of Sampling ▪ Methods of Testing
	IS 417: Part 3: 1986 (Reaffirmed Year: 2017) - Footballs, volleyballs, basketballs, netballs, throw balls and water-polo balls - Part 3 : Basketballs	▪ Dimensions ▪ Other requirements excluding bladders
	IS 10800: 1983 (Reaffirmed Year: 2017) - Cricket Balls	▪ Types of requirements for three grades of balls ▪ Adjunct standards
	IS 4143: 1967 (Reaffirmed Year: 2017) - Carrom draughts	▪ Dimensional and constructional requirements of carrom-draughts. (shape, dimensions and weight) ▪ Moisture content in the type of wood required for making.

Stage	Code	Scope
	IS 3659: 1993 (Reaffirmed Year: 2017) - Table tennis balls-specifications	▪ Requirements and tests for table tennis balls ▪ Adjunct standards ▪ Grades and designations ▪ Manufacture, workmanship and finishes ▪ Sizes

9.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS EN 14468-1-2015** - Table tennis tables, functional and safety requirements, test methods	▪ Lays down functional and safety requirements of table tennis tables ▪ Referenced documents and standards.
	BS EN 14468-2-2015 - Table tennis posts for net assemblies. Requirements and test methods	▪ Lays down functional requirements of table tennis posts for nets. Permanently and temporarily attached. ▪ Referenced documents and standards.

9.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 4905: 2015 (Reaffirmed Year: 2020)** - Methods for random sampling	▪ Random principles of sampling
	IS 4144: 1967 (Reaffirmed Year: 2017) - Carrom strikers	▪ Material and constructional requirements ▪ Adjunct Standards

9.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**BS EN 1270 -** Playing field equipment - Basketball equipment - Functional and safety requirements, test methods	▪ Test methodology for basketball equipment
	ASTM F2117 - 10(2017) - Standard test method for vertical rebound characteristics of sports surface/ball systems; acoustical measurement	▪ Quantitative measurement and normalization of the vertical rebound produced during impacts between athletic balls and athletic surfaces. ▪ Measurement units specification

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information : http://odopup.in/en/article/meerut
2. Product information : https://khelmart.org/
3. Product Information : https://cricketerschoice.com/how-are-cricket-balls-made/
4. Product Information : https://en.wikipedia.org/wiki/Sports_equipment#Game_equipment
5. Product Information : https://www.lords.org/mcc/about-the-laws-of-cricket
6. Product Information: https://www.lords.org/mcc/the-laws-of-cricket/the-ball
7. Product Information : https://www.ittf.com/
8. Product Information : https://www.icfcarrom.com/
9. Product Information: https://kingkongpong.com/what-material-ping-pong-tables-made-of/
10. Product Information: http://www.designlife-cycle.com/basketball
11. Product Information: https://plymouthct.myrec.com/documents/basketball_sizechart.pdf
12. Product Information: https://digitalhub.fifa.com/m/4e93a32cd03542cf/original/testing-manual-fifa-quality-programme-for-footballs-2018.pdf
13. International Legislation/Regulation:
    a. EU: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32010R0995
    b. USA: https://www.cpsc.gov/Business-Manufacturing/Business-Education/Business-Guidance/FHSA-Requirements
    c. AUS: https://forestlegality.org/policy/australia-illegal-logging-prohibition-act
14. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
15. ISO Standards: https://www.iso.org/obp/ui#search
16. International Standards : https://www.lords.org/mcc/the-laws-of-cricket/
17. International Standards : https://www.icfcarrom.com/
18. International Standards : https://www.ittf.com/
19. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB): http://nabcb.qci.org.in/

APPENDIX II: IMPORTANT INSTITUTIONS

SPORTS GOODS				
S. No.	**Institute**	**Areas of Expertise**	**Postal Address**	**Contact**
1	MSME Technology Development Centre, PPDC	▪ Material Testing ▪ Consultancy ▪ Business Incubator	MSME Technology development Centre (Process cum Product Development Centre), Sports Goods Complex, Delhi Road, Meerut, 250002	Email: info@ppdcmeerut.com Phone: 0121-2404991 Website: www.ppdcmeerut.com
2	Forest Research Institute	▪ Forest Conservation ▪ Production and Supply of timber ▪ Planting Stock improvement ▪ Forestry Education ▪ Technology for Eco Friendly Preservatives	Forest Research Institute, Chakarata Rd, New Forest, P.O, Indian Military Academy, Dehradun, Uttarakhand 248006	Email: singhkp@icfre.org Phone: 0135-2224315 Website: https://fri.icfre.gov.in/

ONE DISTRICT ONE PRODUCT

UTTAR PRADESH

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